

ONURCAN YILMAZ

Ankara, Turkey

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SUMMARY

PhD Researcher and Electrical and Electronics Engineer specializing in Control and Robotics, with expertise in ROS, LiDAR-based mapping, odometry, and sensor fusion. Currently extending research toward linear time-periodic (LTP) system identification using Floquet theory and harmonic transfer functions. Experienced in developing real-time robotic systems, hardware/software integration, and advancing robotic perception technologies.

EDUCATION

PhD in Electrical and Electronics Engineering, Hacettepe University Feb 2025 - ongoing

Research Area: LTP System Identification and Control

Funded by TÜBİTAK 2211-A National PhD Scholarship Programme

MSc in Electrical and Electronics Engineering, Hacettepe University (GPA: 3.58) Feb 2022 - Feb 2025

Funded by 5G and Beyond Joint Graduate Support Program and TÜBİTAK 2210-A

Thesis: *Sensor Fusion and Robotic Mapping using 3D LiDAR Odometry on a Quadrupeled Robot Platform*

BSc in Electrical and Electronics Engineering, Middle East Technical University 2015 - Feb 2021

Graduation Project: *Design and Construction of a Robot that can make Mapping and Localization*

Member and 1-term Vice President, IEEE METU Robotics and Automation Society

AWARDS

- **2nd Place, Robotics, Control & Automation Track** — 34th Signal Processing and Communications Applications Conference (SIU), 2026.
- **TÜBİTAK 2211-A National PhD Scholarship Programme** — The Scientific and Technological Research Council of Turkey (TÜBİTAK), 2025.
- **TÜBİTAK 2210-A National MSc Scholarship Programme** — The Scientific and Technological Research Council of Turkey (TÜBİTAK), 2022.
- **5G and Beyond Joint Postgraduate Support Program** — BTK, in partnership with METU, Hacettepe University & Bilkent University, 2022. Included a funded Research Engineer appointment (5G VATS R&D Associate, Turkcell Technology, May 2022 – Dec 2023).

THESIS

- **Onurcan Yılmaz**, *Sensor Fusion and Mapping on a 4 Legged Robot System with 3D LiDAR*, M.Sc. Thesis, Hacettepe Üniversitesi, Ankara, Türkiye, February 2025. Advisor: Doç. Dr. İsmail Uyanık.

JOURNAL ARTICLES

- **Onurcan Yılmaz**, Eren Cem Göksülük, İsmail Uyanık, *Sistem tanılama yöntemlerinin zayıf elektrik balıklarının hareket dinamikleri üzerinden karşılaştırılması*, Journal of the Faculty of Engineering and Architecture of Gazi University, Vol. 40, No. 2, pp. 877–887, 2025.

CONFERENCE PAPERS

- **Onurcan Yılmaz**, İsmail Uyanık, *Floquet-Based Periodic Matrix Reconstruction in Discrete-Time LTP Systems*, 2026 34th Signal Processing and Communications Applications Conference (SIU), İstanbul, Türkiye, 2026. 2nd Place, Robotics, Control & Automation Track. DOI: 10.1109/SIU71813.2026.11637048
- Emin Yusuf Aydın, **Onurcan Yılmaz**, İsmail Uyanık, Mustafa Öztürk, *Structure-Aware Filtering for Real-Time Separation of Active-Sensing and Tracking Motions*, 2026 34th Signal Processing and Communications Applications Conference (SIU), İstanbul, Türkiye, 2026. DOI: 10.1109/SIU71813.2026.11636672

- **Onurcan Yılmaz**, İsmail Uyanık, *Improving LiDAR Odometry with Hausdorff Distance-Based Variance Estimation*, 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE), pp. 2477–2482, 2025. DOI: 10.1109/CASE58245.2025.11163897

EXPERIENCE

Research and Teaching Assistant

Dec 2023 - ongoing

Hacettepe University, Department of Electrical and Electronics Engineering

Ankara

- Investigating linear time-periodic (LTP) system identification, Floquet theory and harmonic transfer functions.
- Conducting research on robotic mapping, sensor fusion, and SLAM algorithms.
- Conducting undergraduate laboratory sessions across multiple EEE courses. (2023 – ongoing)
- Mentoring senior-year graduation projects related to robotics. (2023 – ongoing)

5G VATS R&D Associate

May 2022 - Dec 2023

Turkcell Technology

Ankara

- Integrated VLP-16 LiDAR and ZED Stereo Camera on a quadruped robot with Jetson NX.
- Developed covariance estimation methods for LiDAR Odometry algorithms.
- Developed a ROS driver for ArNav S-series IMU, implementing a custom binary serial protocol with sensor data parsing and ROS message publishing. (2023)

Hardware Design Engineer

Mar 2021 - May 2022

Kuartis Technology

Ankara

- Designed STM32-based embedded systems hardware using Altium Designer.
- Developed embedded software using FreeRTOS.
- Built automated testing tools using Raspberry Pi for ESP32-based products.
- Oversaw MIL-STD-810-H compliance tests.

Intern

2019 Summer

Garanti Technology

Istanbul

- Tested new released z/OS migration on a sandbox environment.
- Assisted in system maintenance and troubleshooting tasks.

Intern

2018 Summer

SPACE & DEFENCE TECHNOLOGIES (SDT)

Ankara

- Worked on embedded systems software development.
- Implemented sensor data acquisition and cloud storage using MQTT protocol.

SKILLS

Programming	C/C++, Python, MATLAB (Simulink)
Robotics	ROS, SLAM, LiDAR Odometry
Control & Systems	LTP System Identification, Floquet Theory, Harmonic Transfer Functions
Embedded Systems	STM32, FreeRTOS
Hardware Design	Altium Designer, Fusion 360
Tools	Git, Linux, Jetson Platforms
Languages	Turkish (Native), English (Professional)

PROJECTS

Graduation Project - Design and Construction of a Robot that can make Mapping and Localization

Design and construct a vehicle which can extract the physical and magnetic map of a 1.5m x 2.5m game field containing physical and magnetic landmarks. Once the map is extracted, the vehicle is required to localize itself along its path based on the extracted map. The vehicle should be remotely controlled. [Final Presentation](#)

EEE STAR Project - Orangutan - Passive Dynamic Walker Reconstruction of a robot named [Rando](#) with 2 new IMU sensors and Wi-Fi connection to have online angular velocity measurements of the legs for loop closure. For more information, you can see our [Poster](#) at the project fair.

BallBot Construction of a Ballbot which can balance and move over a ball. Worked on hardware and software development of torque control of a DC motor. Conducted with IEEE METU RAS.

EXTRA-CURRICULAR ACTIVITIES

- Camping, Swimming, Snorkelling, Diving (SSI AOWD) and occasionally Spearfishing

LEADERSHIP

Vice President of IEEE METU Robotics and Automation Society

2018 - 2019

- Coordinated technical affairs of the society like; Projects, Workshops, etc.